Eccentric Eccentricities

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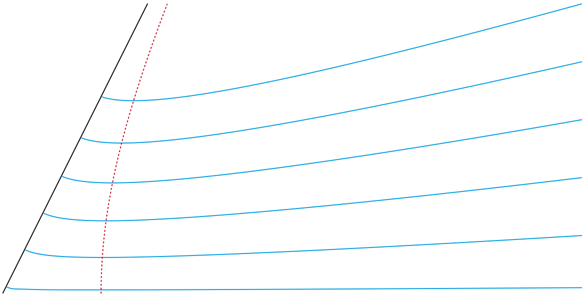

Theoretical Developments in Gravitational Wave Physics, YITP, 17 June 2026

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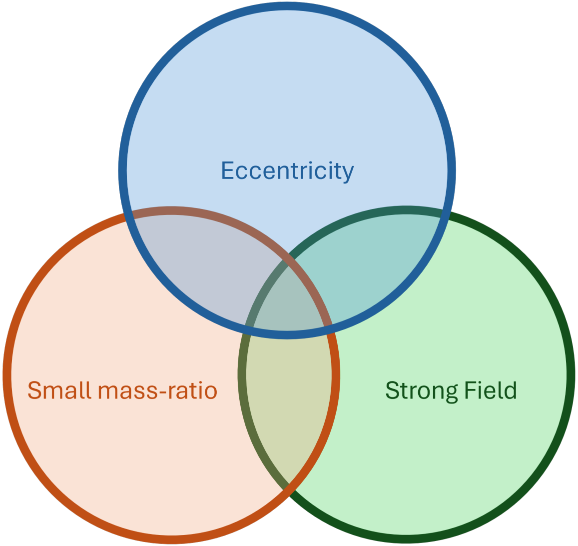
Context



"Emission of gravitational waves causes binaries to become more circular over time."

True in weak field approximations:

Quadrupole approximation [Peters,1964]

$$\dot{e} = -\frac{304}{15} e \frac{G^3 M^3 \nu}{c^5 p^4} (1 - e^2)^{3/2} \left(1 + \frac{121}{304} e^2\right)$$

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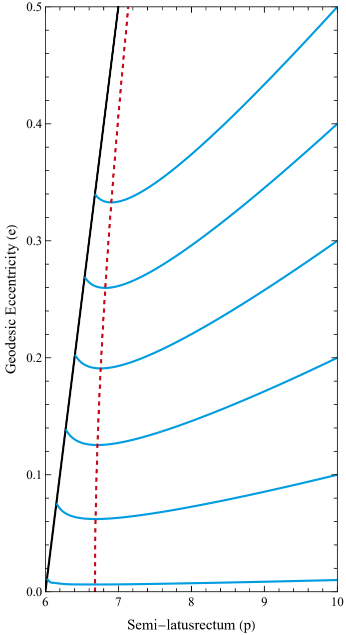
Exception



In the final stage of extreme mass-ratio inspirals eccenticity can increase.

- First described: [Apostolatos+,1993][Tanaka+,1993]
- Well-documented: [Cutler+,1994][Glampedakic&Kennefick,2003] [Warburton+,2011][MvdM+,2018][Lynch+,2022]
- Zoom-Whirl regime ($n_{\rm wind} \approx 3$)
- Leading order (0PA) effect
- $\dot{e} = 0$ to be used as reference point by LISA SGS

Questions:

- Is this physical?
- Is this expected?
- Is this a unique feature of GR?

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What is eccentricity anyway?

Rough definition:

Eccentricity measures how non-circular a trajectory is.

Desirable qualitities of a good definition of eccentricity:

1. (Quasi-)circular trajectories have $e = 0$
2. Bound trajectories have $e < 1$
3. Scattering trajectories have $e > 1$
4. Different representations of the same trajectory should agree on e
5. Agrees with Newtonian limit e

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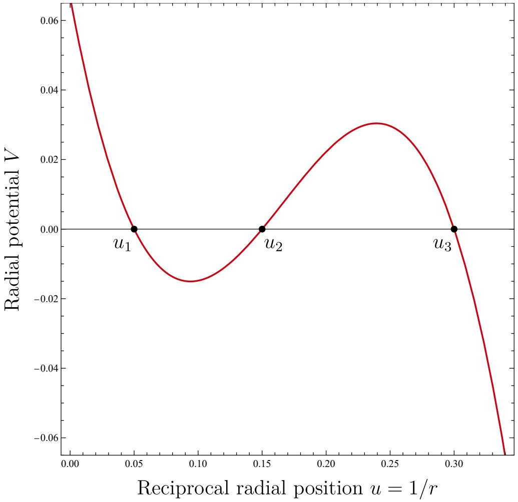
$$u = M/r$$

$$\frac{1}{2}\dot{u}^2 + V[\mathcal{E}, \mathcal{L}](u) = 0$$

$$e_{\text{geo}} = \frac{u_2 - u_1}{u_2 + u_1}$$

Desirable qualitities of eccentricity:

1. (Quasi-)circular trajectories have $e = 0$✓
2. Bound trajectories have $e < 1$ ✓
3. Unbound trajectories have $e > 1$ ✓
4. Different representations agree on e ✗
5. Agrees with Newtonian limit e ✓



GW eccentricity

$$\omega_{22} = \frac{\mathrm{d}}{\mathrm{d}t} \arg h_{22}$$

$$e_{\omega_{22}} = \frac{\sqrt{\omega_{22}^{p}} - \sqrt{\omega_{22}^{a}}}{\sqrt{\omega_{22}^{p}} + \sqrt{\omega_{22}^{a}}}$$

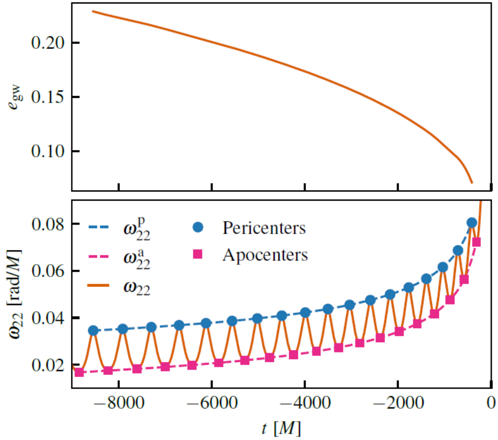
$$e_{\mathrm{GW}} = \cos(\Psi/3) - \sqrt{3}\sin(\Psi/3)$$

$$\Psi = \arctan\left(\frac{1 - e_{\omega_{22}}^2}{2 e_{\omega_{22}}}\right)$$

[Ramos-Buades+MvdM,2022]

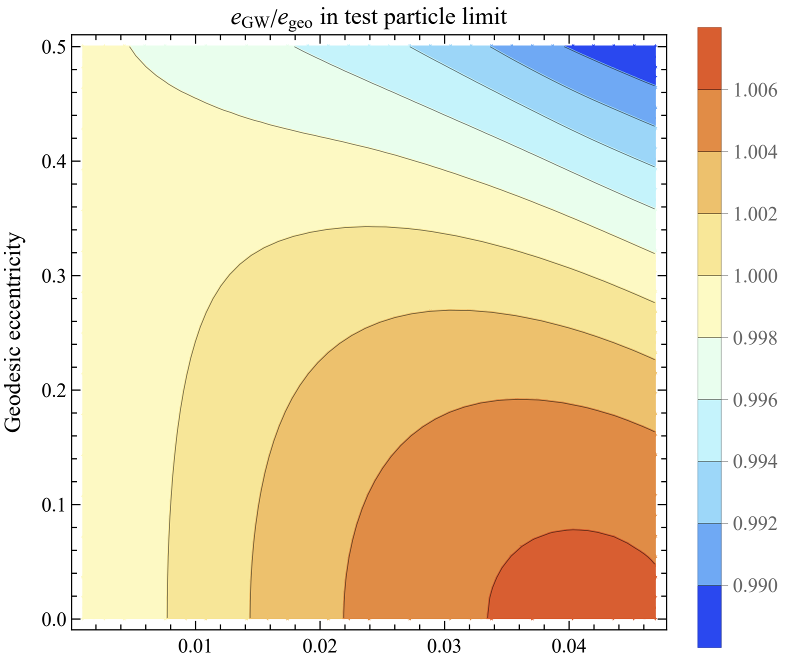
Desirable qualities of eccentricity:

1. (Quasi-)circular trajectories have $e = 0$ ✓
2. Bound trajectories have $e < 1$ ✓
3. Unbound trajectories have $e > 1$ ✗?
4. Different representations agree on e ✓*
5. Agrees with Newtonian limit e ✓



[Shaikh+MvdM,2023]

GW and geodesic eccentricity are similar in small mass-ratio limit



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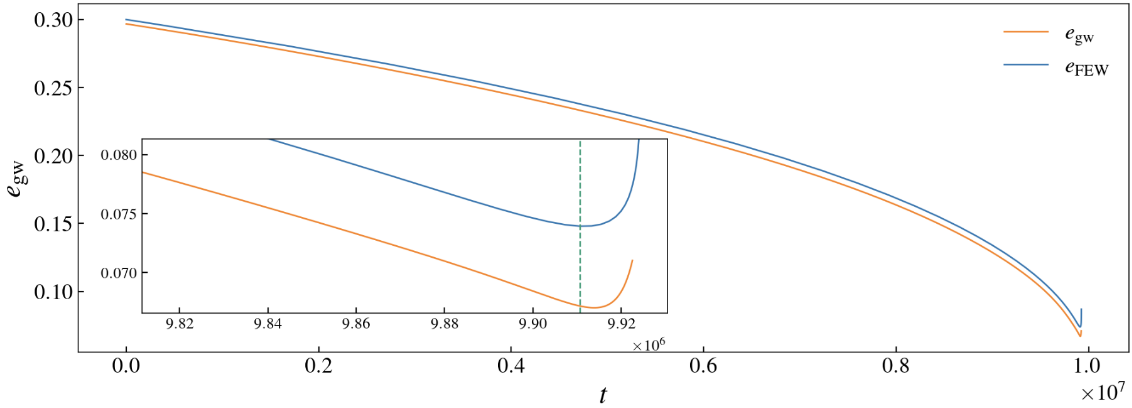
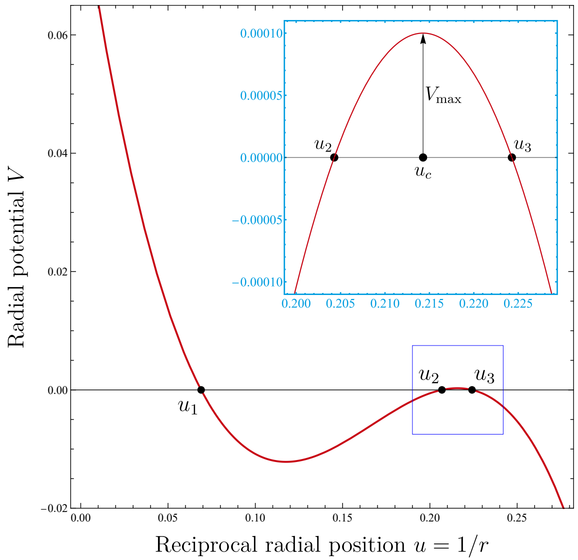
$m_1 : m_2 = 10^4 : 1$



[Courtesy Arif Shaikh]



Transition from stable bound orbit to plunge happens when $u_2 = u_3$.

Near the critical point:

$$V = AV_{\max} - A(u - u_c)^2 + \mathcal{O}(u - u_c)^3$$

(If $V[\mathcal{E}, \mathcal{L}]$is smooth then so are $A[\mathcal{E}, \mathcal{L}]$, $V_{\max}[\mathcal{E}, \mathcal{L}]$, and $r_c[\mathcal{E}, \mathcal{L}]$)

Pericenter can be approximated:

$$u_2 \approx u_c - \sqrt{V_{\max}}$$

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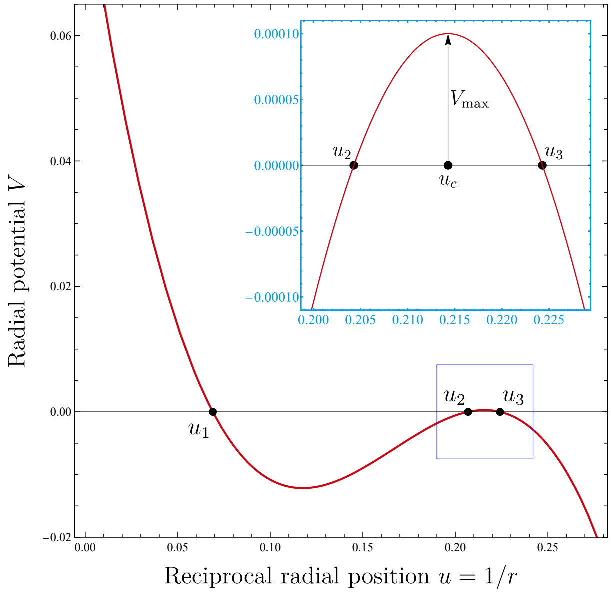

$$\dot{e} = \frac{2u_1}{(u_1+u_2)^2}\dot{u}_2 - \frac{2u_2}{(u_1+u_2)^2}\dot{u}_1$$

$$\dot{u}_2 \approx \dot{u}_c - \frac{\dot{V}_{\max}}{2\sqrt{V_{\max}}} \approx \dot{u}_c - \frac{\dot{V}_{\max}}{2(u_c - u_2)}$$

Conclusion

$\dot{e}$ diverges to $+\infty$ if $\dot{V}_{\max} < 0$ and does not approach 0.

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Orbits near the LSO (Zoom-Whirl)

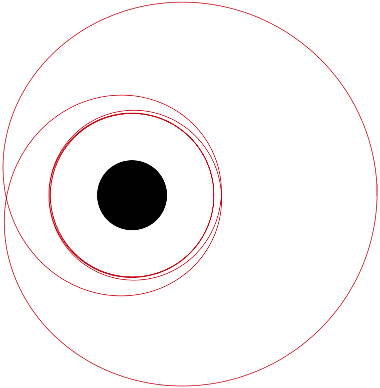
Time spent near pericenter

$$2\int_{u_2-\epsilon}^{u_2} \frac{\mathrm{d}t}{\mathrm{d}u}\,\mathrm{d}u \propto \int_{u_2-\epsilon}^{u_2} \frac{1}{\sqrt{(u-u_c)^2 - V_{\max}}}\,\mathrm{d}u$$
$$\approx \int_{u_2-\epsilon}^{u_2} \frac{1}{u_c - u}\,\mathrm{d}u \approx \log(u_c - u_2)$$

$n_{\text{wind}} = \frac{\Omega_\phi}{\Omega_{\text{rad}}}$: Winding number

$$n_{\text{wind}} \stackrel{?}{=} n_{\text{whirl}} + n_{\text{zoom}}$$

$n_{\text{zoom}} \approx 1$ and $n_{\text{whirl}} \sim \log(u_c - u_2)$



For $\mathcal{C} = \mathcal{E}, \mathcal{L}$

($\Delta\mathcal{C}$ in radial period) $= n_{\text{whirl}} \times$ ($\Delta\mathcal{C}$ in circular orbit at u_c) + (finite $\Delta\mathcal{C}$)

After dividing by the radial period (to obtain average):

$$\dot{\mathcal{C}} \approx \dot{\mathcal{C}}_{\text{UCO}} + \frac{f(p,e)}{\log(u_c - u_2)}$$

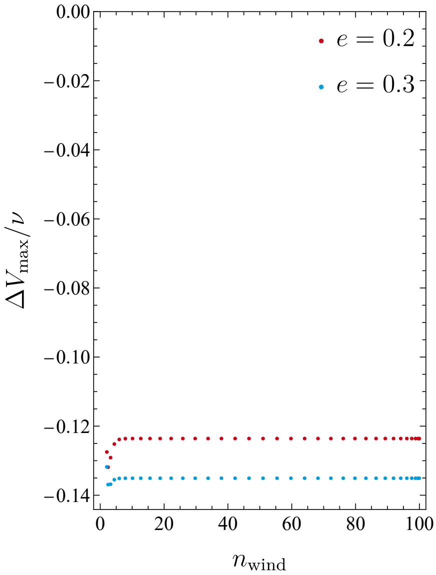
Evolution of the peak of the potential

$$\dot{V}_{\max} = \frac{\partial V_{\max}}{\partial \mathcal{E}}\dot{\mathcal{E}} + \frac{\partial V_{\max}}{\partial \mathcal{L}}\dot{\mathcal{L}}$$

On unstable circular orbits (in GR) $\dot{V}_{\max} = 0$

$$\dot{V}_{\max} \approx 0 + \frac{f(u_c - u_2)}{\log(u_c - u_2)}$$

In Schwarzschild:



[Data: Lucía Vélez]

Back to $\dot{e}$

$$\dot{e} \sim \frac{-1}{(u_c - u_2)\log(u_c - u_2)}$$

Eccentricity will always increase before LSO transition.

Ingredients

- Transition through LSO is a double root.
- (Radiation reaction maintaines circularity)
- Evolution is sufficiently adiabatic.

~~Questions~~ Answers:

- Are eccentricity upticks physical? YES
- Are eccentricity upticks expected? YES
- Are eccentricity upticks unique to GR? NO

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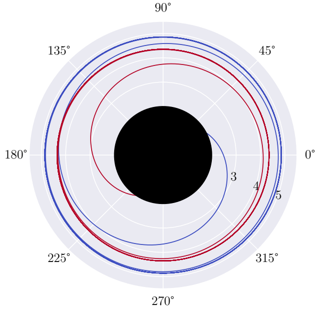
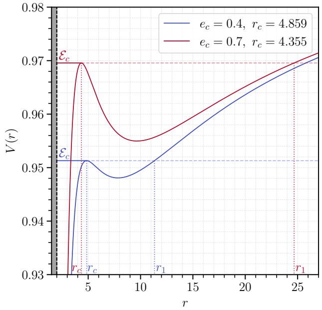
"The peak of the waveform is associated with merger"





[Faggioli+ (MvdM), 2025]

- $\nu \to 0$ limit for EMRI plunge.
- Can be expressed in elementary functions [Dyson&MvdM, 2023]
- Calculate waveform with TD Teukolsky code.

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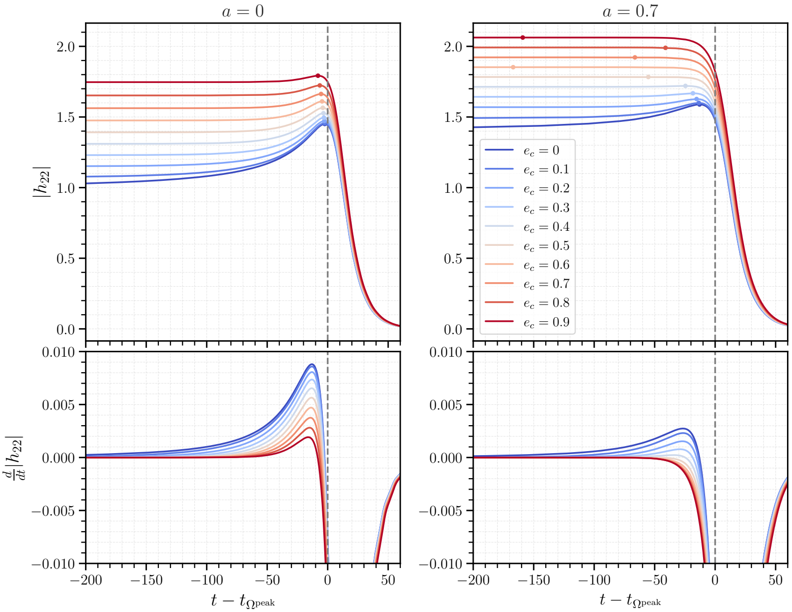
Critical Plunge Geodesics



[Faggioli+ (MvdM), 2025]

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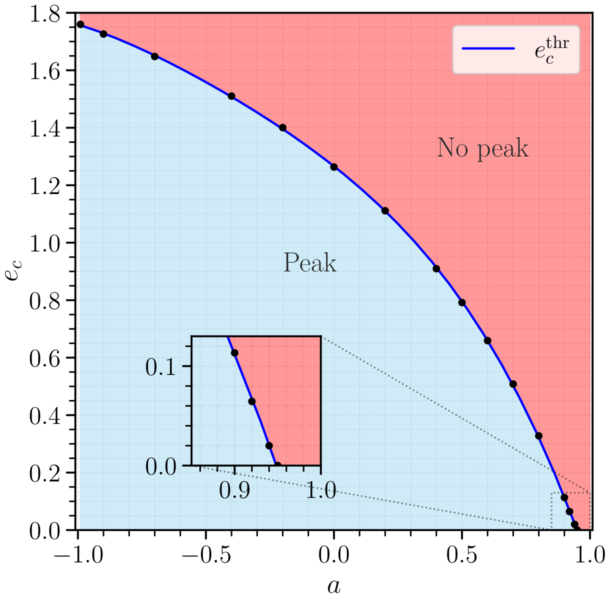

- QC case: [Tarachini+, 2014]
- Near-extremal spin, [Warburton+, 2016]

[Faggioli+ (MvdM), 2025]

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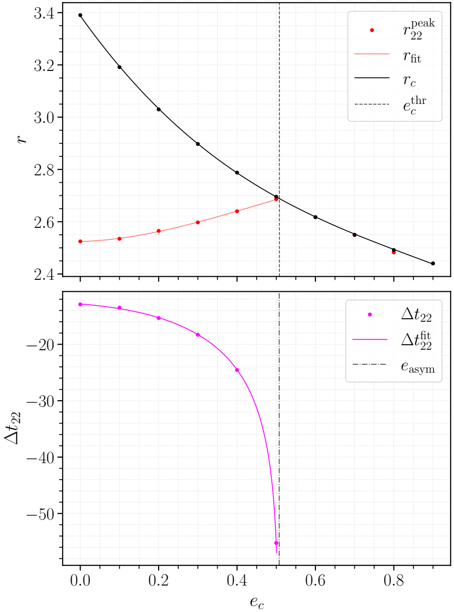
Explanation



[Faggioli+ (MvdM), 2025]

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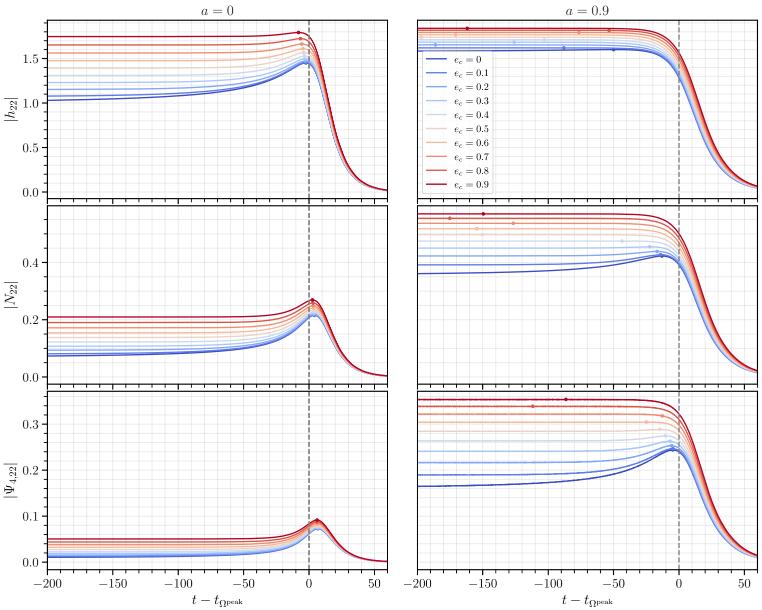
Not just the strain



[Faggioli+ (MvdM), 2025]

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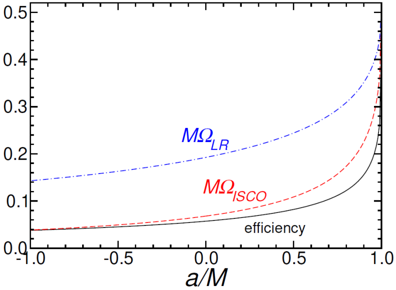
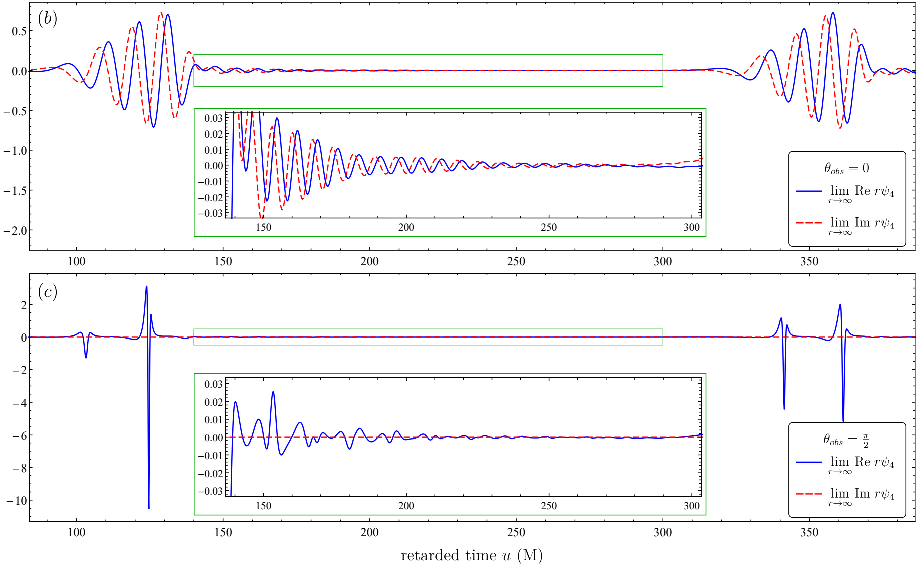
"Inspirals do not excited QNMs"



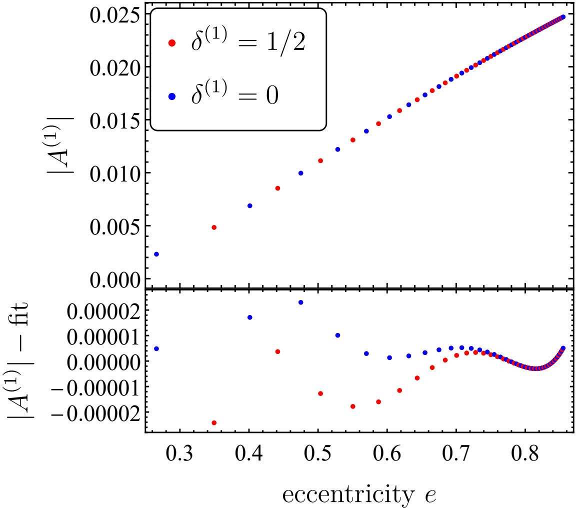
[Berti+, 2009]

ψ_4 at $\mathcal{I}^+$ for $a = 0.99$, $p = 3$, $e = 0.8$



Explanation

- Eccentricity broadens spectrum
- Direct radiation alternates between strong high frequency and weak low frequency
- Relatively weak effect
- Only mildly sensitive to QNM resonances



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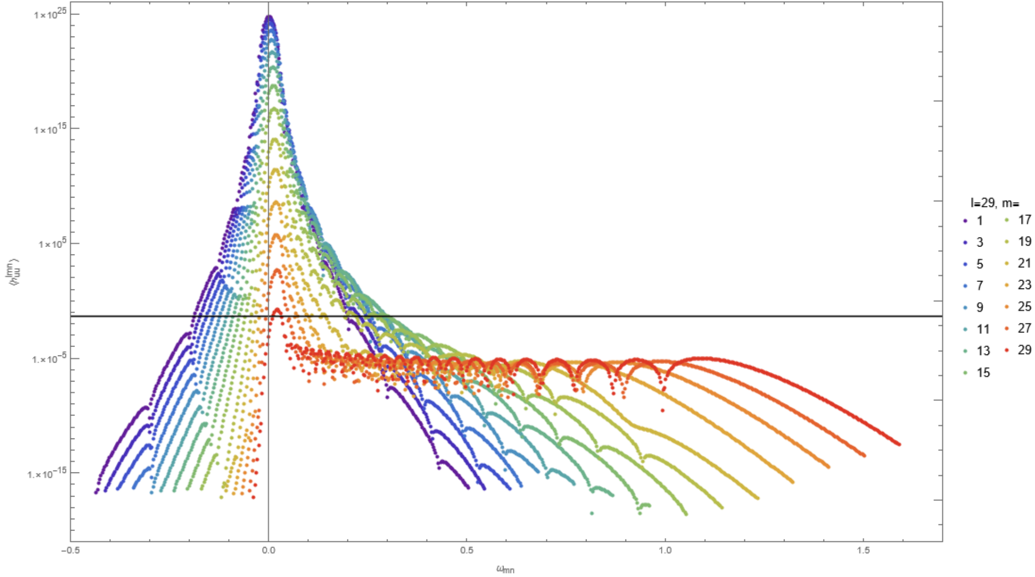
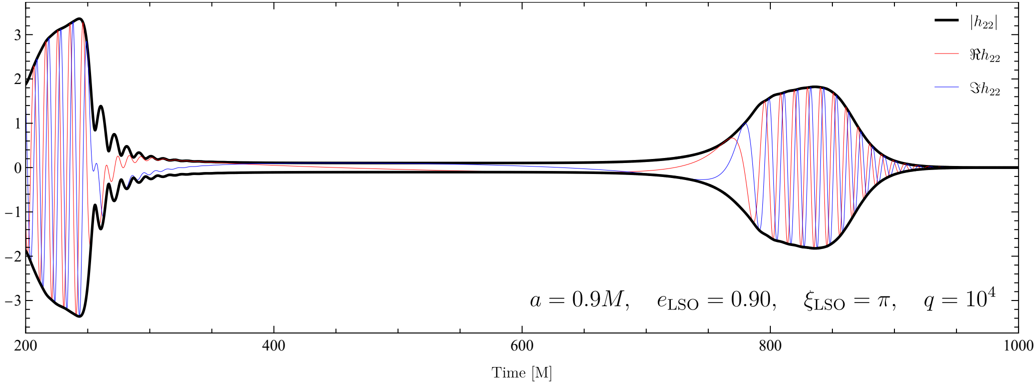
$\ell = 29$ contributions to 1SF redshift invariant for $a = 0$, $p = 16.2$, $e = 0.8$



Revenge of the Wiggles II [Faggioli+ (MvdM), 2026]



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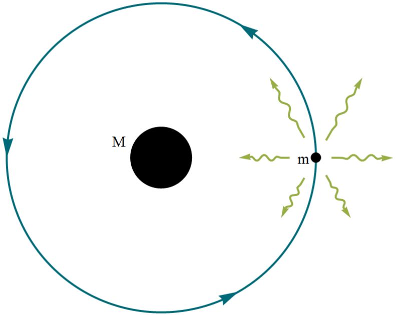
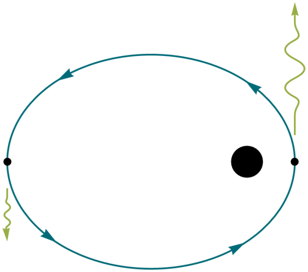
"EMRI velocity kicks scale with ν^2"

Radiation inbalance

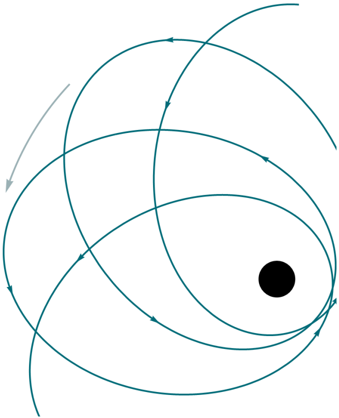


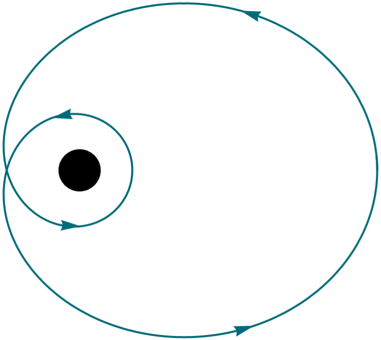
- Beaming causes radiation inbalance $a \propto \nu^2$
- Effect average to 0 over orbit.
- On inspiral timescale "kick" $\Delta v \propto \nu^2$

Eccenrtic orbits

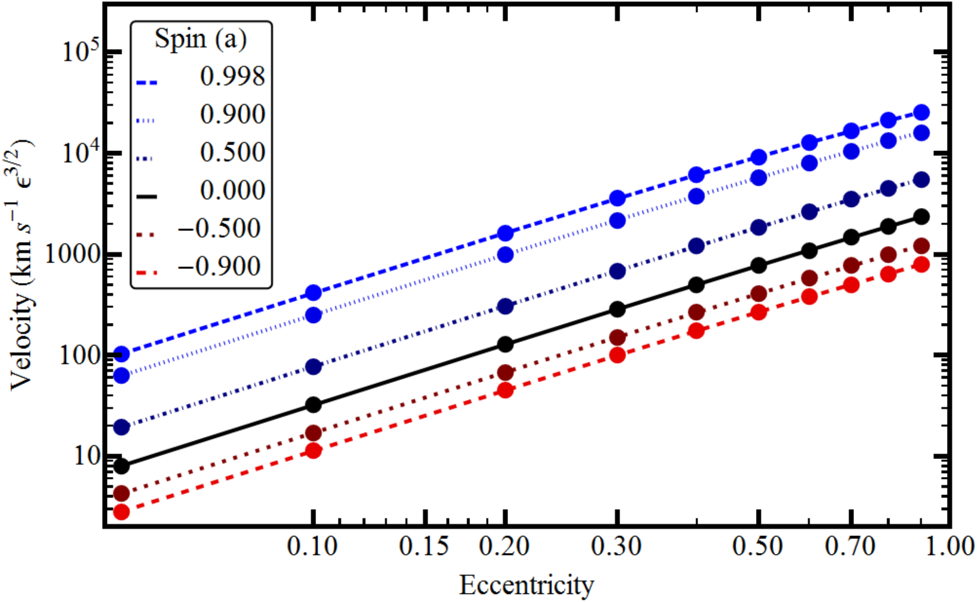


- Inbalance between pericenter and apocenter causes $\Delta v_{orb} \propto \nu^2$
- Effect average to 0 due to pericenter precession.
- On inspiral timescale “kick” $\Delta v_{insp} \propto \nu^2$





- Pericenter advance $0 \mod 2\pi$
- "Kick" builds coherently over time $\propto \frac{1}{\sqrt{\nu}}$
- Resonantly enhanced "kick" $\Delta v_{res} \propto \nu^{3/2}$



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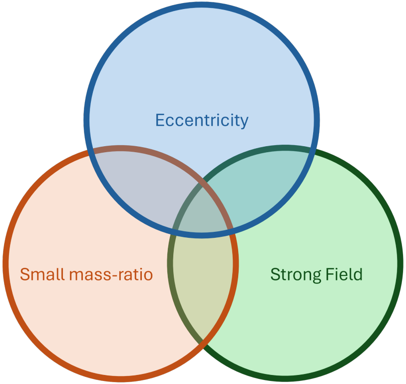

Eccentricity 1:

Eccentricity will grow at the end of inspiral (for $\nu \lesssim 10^{-3}$)

Eccentricity 2:

The waveform can peak long before merger.

Eccentricity 3:

Eccentric orbits will excite QNMs during inspiral.

Eccentricity 4:

Eccentric inspirals experience CoM kick $\propto \nu^{3/2}$